

Problem

The current entering the positive terminal of a device is $i(t) = 6e^{-2t}$ mA and the voltage across the device is $v(t) = 10 \frac{di}{dt}$ V.

(a) Find the charge delivered to the device between $t = 0$ and $t = 2$ s.

(b) Calculate the power absorbed.

(c) Determine the energy absorbed in 3 s.

Step-by-step solution

Step 1 of 3

The current entering the positive terminal of a device $i(t) = 6e^{-2t}$ mA

The voltage across the device $v(t) = 10 \frac{di}{dt}$ V

(a)

Calculate the charge delivered to the device between $t = 0$ and $t = 2$ s.

$$\begin{aligned} q &= \int_0^2 i(t) dt \\ &= \int_0^2 (6e^{-2t} \times 10^{-3} \text{ A}) dt \\ &= (6 \times 10^{-3}) \left(\frac{e^{-2t}}{-2} \right) \Big|_0^2 \end{aligned}$$

Substituting the values, we get,

$$\begin{aligned} q &= -(3 \times 10^{-3}) (e^{-4} - e^{-0}) \\ &= -(3 \times 10^{-3}) (0.0183 - 1) \\ &= 2.945 \text{ mC} \end{aligned}$$

Therefore, the charge q delivered to the device $t = 0$ and $t = 2$ s is $\boxed{2.945 \text{ mC}}$.

Step 2 of 3

(b)

The power absorbed by the device is:

$$\begin{aligned} p &= v(t) i(t) \\ &= \left(10 \frac{di}{dt} \text{ V} \right) (6e^{-2t} \times 10^{-3} \text{ A}) \\ &= \left(10 \frac{d}{dt} (6e^{-2t} \times 10^{-3} \text{ A}) \text{ V} \right) (6e^{-2t} \times 10^{-3} \text{ A}) \\ &= (-120e^{-2t} \times 10^{-3} \text{ V}) (6e^{-2t} \times 10^{-3} \text{ A}) \\ &= -720e^{-4t} \mu\text{W} \end{aligned}$$

Therefore the power p consumed by the device is $\boxed{-720e^{-4t} \mu\text{W}}$.

Step 3 of 3

(c)

The energy absorbed in 3s can be calculated as:

$$\begin{aligned}w &= \int_0^3 p \, dt \\&= \int_0^3 (-720e^{-4t} \, \mu\text{W}) \, dt \\&= -720 \left(\frac{e^{-4t}}{-4} \right)_0^3 \\&= 180 \times 10^{-6} (e^{-12} - e^{-0}) \\&= 180 \times 10^{-6} (6.15 \times 10^{-6} - 1) \\&= -180 \, \mu\text{J}\end{aligned}$$

Since $6.15 \times 10^{-6} \ll 1$, $6.15 \times 10^{-6} - 1$ is taken as -1 .

Therefore, the energy w absorbed in 3s is $\boxed{-180 \mu\text{J}}$.